CryptoLens: An Interactive Visual Analytics System for Financial Market Intelligence

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1 Introduction

Financial markets generate millions of data points every second. This includes **price movements**, **queued orders**, **trading volumes**, and the **relationships between different assets**. Even with all this rich information, most investors still just look at **static line charts** of past closing prices. This hides the actual **dynamic structure of the market**. It makes it hard to see where prices might hit **resistance**, how **risk** changes over time, whether a past pattern points to a recovery or a drop, and how various assets move together.

CryptoLens fixes this problem by offering an **interactive visual analytics system** that fits entirely on a **single screen**. It runs on five years of data captured down to the minute for five major cryptocurrencies: **Bitcoin (BTC)**, **Ethereum (ETH)**, **Binance Coin (BNB)**, **Solana (SOL)**, and **XRP**. In total, that adds up to roughly **10.5 million records**.

The system is a **web application** that displays all five analytical views at the same time. An **asset selector** is available to switch the **active cryptocurrency**, updating every view instantly. These five views handle **timeline exploration**, **live market depth monitoring**, **hypothesis testing** on past price patterns, **risk quantification** based on volatility, and **correlation analysis** between different assets. Everything is **linked together interactively**, meaning a selection made in one view immediately updates the rest.

2 Data Source

Binance Historical Data Portal

Binance offers an official public data repository at **data.binance.vision**. It provides complete **one minute candlestick data** for all trading pairs. This includes the **open price**, **high price**, **low price**, **close price**, and **trading volume**. The platform requires **no API key** or any kind of **authentication**. Monthly data files are downloadable directly via a web browser using URL'S of the form :

https://data.binance.vision/data/spot/monthly/klines/BTCUSDT/1m/BTCUSDT-1m-2022-01.zip

The above link shows minute level BTC(bitcoin) data of January 2022 in which each row denotes the following values:-

OPEN TIME	OPEN	HIGH	LOW	CLOSE	VOLUME	TRADES	BUY VOLUME
1499040000000	$42,150.00	$42,390.00	$42,080.00	$42,310.00	148,976 BTC	308	89,420 BTC

All available files can be viewed at their public directory, which is completely open and requires **no login** whatsoever. Extracting the data for five assets from January 2020 through December 2024 yields approximately **10.5 million rows** of **structured time series data**. This dataset is also accessible programmatically using the **binance historical data Python package**.

Live Order Book Data

For live order book data, Binance streams real time information through their public **WebSocket API**. A **WebSocket** is a protocol for a continuous **live data stream**. Unlike a regular website link, it cannot be opened directly in a browser and must be accessed **programmatically**. The stream endpoint :-

wss://stream.binance.com:9443/ws/btcusdt@depth20@100ms

delivers the **top 20 bid and ask levels**. This reveals the queued **buy and sell orders** at every price point, updating every **100 milliseconds** without needing any login credentials. **Live order book data** is easily accessible in real time by running a Python script connected to the public WebSocket. Meanwhile, all **historical candlestick data** remains available to download straight from the vision portal entirely hassle free.

3 Specific Tasks

- **Data Ingestion and Pipeline:** The full dataset from the Binance data portal will be downloaded and stored into a **PostgreSQL database** optimized for **time series data** using automated **Python scripts**.
- **Data Preprocessing and Feature Engineering:** Derived columns such as **percentage returns**, **rolling standard deviations**, and **moving averages** will be computed. Furthermore, minute data will be aggregated into **hourly and daily resolutions** using **Pandas resample operations**.
- **Real Time Streaming:** A **WebSocket relay server** will be constructed to ingest live order book data from Binance and forward it to the frontend with **sub second latency**.
- **Visual Analytics Development:** Five **interactive coordinated views** featuring **brushing and linking** will be implemented using **D3.js** and **Plotly**. This structure ensures all views respond seamlessly to selections made in any single view.
- **Statistical Analysis:** The methodology involves **pattern scanning** across 10.5 million rows, **volatility regime detection**, **return distribution modelling** under normal and stress conditions, alongside **Pearson correlation matrix** computation utilizing **rolling windows**.

4 Visualisation Tasks

4.1 The Time Machine : Multi-Resolution Temporal Explorer

A compressed multi-year price overview at the top of the screen allows users to drag-select any time window, which is a technique formally known as **"brush and link"**. All other views instantly update to show only the selected period. The detail panel renders the **candlestick charts**, visual bars showing the **open, high, low, and close price** for each time period with a synchronised **volume panel** below showing trading activity. Resolution adapts automatically to the selected window size: **daily candles** for multi-month windows, **hourly** for multi-week, and raw **one-minute candles** for short windows.

Real-world value: analysts need to examine the same market event at multiple time scales simultaneously without losing the broader context.

4.2 Market Depth Visualiser : Order Book Analytics

The **order book** is the live record of all **pending buy orders** below the current price and **sell orders** above it, across all price levels. This feature renders it as a **mirrored area chart**: cumulative **buyer volume** rises toward the current price on the left side, and cumulative **seller volume** rises away from it on the right. Sudden steep jumps in either curve, which are known as **support walls** (buyer side) and **resistance walls** (seller side), identify price levels where the market is likely to slow or reverse. In live mode, data streams via **WebSocket** at **100ms intervals**; in historical mode, depth is reconstructed from **OHLCV data** for analysis of past **market structure**.

Real-world value: identifying price walls before placing large orders prevents unfavourable trade execution.

4.3 The What-If Simulator : Pattern Hypothesis Tester

Traders often act on hunches, such as "every time the price drops sharply in an hour, it tends to recover the next day." The **What-If Simulator** allows for testing whether such a hunch is **statistically supported** or just a gut feeling. A simple **trigger condition** based on **price movement** or **volume behaviour** is specified. The system then scans all **historical records**, finds every past instance where that condition occurred, and visualises what happened to the price after each such event, such as whether it rose, fell, or stayed flat. This provides **evidence-based insight** into whether a pattern is real and repeatable, or coincidental.

Real-world value: replacing intuition with data-driven validation of trading hypotheses.

4.4 Volatility Engine : Market Risk Analytics

Volatility measures how unpredictably and widely a cryptocurrency's price swings over time. A **high-volatility** asset can gain or lose 15% in a single day, while a **low-volatility** one moves within a narrow range. This feature visualises how **market risk** evolves across the selected time window and models the **statistical probability** of large price movements under three conditions: 1) **normal market behaviour**, 2) **market stress**, and 3) a **crash event**. These three conditions can be switched interactively to observe how the probability of a significant price drop changes under each scenario.

Real-world value: understanding tail risk — the probability of extreme adverse price movements — is fundamental to risk management in any financial institution.

4.5 Cross-Asset Correlation Matrix

Correlation measures how consistently two cryptocurrencies move in the same direction. If Bitcoin and Ethereum almost always rise and fall together, they are **highly correlated**, meaning holding both provides little **diversification benefit** since they tend to crash simultaneously. This feature displays an **interactive heatmap** showing the correlation between every pair of the five assets for the currently selected time window. Any pair can be clicked to drill down and see exactly how the two assets moved relative to each other over time, and how their relationship has strengthened or weakened across different **market conditions**.

Real-world value: correlation analysis is a core tool in portfolio construction and risk management.

5 Overall Solution

CryptoLens is a **web application** built entirely on a **single page**. It shows all five analytical views at once on a **fixed screen**, eliminating the need to scroll. An **asset selector** at the top allows for changing the **active cryptocurrency**, which updates every view instantly. The only exception is the **Cross-Asset Correlation Matrix**, which always shows all five coins together regardless of the selection. Everything is deeply connected behind the scenes. If a specific time window is selected in the **Time Machine**, that choice immediately updates the **Volatility Engine**, the **Correlation Matrix**, and the **What-If Simulator**. On the backend, **REST API endpoints** are utilized to pull in **OHLCV data** at various resolutions, alongside a **WebSocket connection** to stream **live order book data**. The heavy lifting is kept on the **server side** to handle all the complex **statistical math**. That frees up the **frontend** to focus entirely on rendering the graphics and maintaining a smooth, **interactive experience**.

6 Tech Stack

Layer	Technology
Data Ingestion	Python, Pandas, binance-historical-data
Data Processing	Python, Pandas, NumPy, SciPy
Database	PostgreSQL / TimescaleDB
Backend REST API	Python, FastAPI
WebSocket Server	Node.js, Express, ws library
Frontend	React.js
Charting	D3.js, Plotly.js
Version Control	GitHub

7 Team Members and Responsibilities

Member (Roll No.)	Primary Responsibility
Soumya Jhunjhunwala (241034)	Data pipeline: Binance ingestion scripts, database schema, automated download
Vivek Singh (241186)	Backend API: REST endpoints for OHLCV aggregation (FastAPI)
Arpit Kumar Singh (240194)	Real-time infrastructure: WebSocket relay server, live order book pipeline
Aman (240097)	Frontend architecture: React app structure, state management, API integration
Pratyush Sandhwar(240788)	Feature 1 — Time Machine: canvas rendering, brush interaction, adaptive zoom
Lakshya Arukiya (240589)	Feature 2 — Order Book Visualiser: live depth chart, wall detection, hover UI
Aadi Sehal (240005)	Feature 3 — What-If Simulator: pattern scanner, outcome visualisation
Anandan Iyer (240117)	Feature 4 & 5 — Volatility Engine and Correlation Matrix: stats + visualisation
Tulika (241106)	Integration, testing, GitHub repository management and final demo setup